

NETWORK FORENSICS INVESTIGATION REPORT

SBT-DF203: Basic Networking Skills for Digital Forensics
Lab 3 — SYN Flood Pattern Investigation Using TShark

Report Reference	SBT-DF203-Lab3-2025FWSD11459-AdeleyeAdemiluyi
Course	SBT-DF203: Basic Networking Skills for Digital Forensics
Lab Number	Lab 3 — SYN Flood Pattern Investigation Using TShark
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Institution	International Cybersecurity and Digital Forensics Academy (ICDFA)
Registration No.	2025/FWSD/11459
Analysis Platform	Kali Linux VM — Apache2, tshark, Scapy
Date of Analysis	2026-09-15

Safety Declaration: Simulation limited to exactly 4 SYN packets sent to 127.0.0.1:80 only. No external systems targeted.

1. Executive Summary

This report documents a network forensic investigation into TCP SYN flood activity against a training Apache web service as part of ICDFA SBT-DF203 Lab 3. Three captures were analysed: a normal HTTP baseline, a bounded 4-packet loopback simulation using Scapy, and a supplied training PCAP (mySYNFloodCapture.pcap). Key SYN flood indicators were identified including incomplete TCP handshakes, randomised source ports, absence of HTTP application data and oversized SYN packets in the supplied PCAP.

2. Evidence and Chain of Custody

Field	Value
Case Identifier	SBT-DF203-Lab3-2025FWSD11459-AdeleyeAdemiluyi
Trainee	Adeleye Ademiluyi Timilehin
Evidence 1	mySYNFloodCapture.pcap — supplied training PCAP (255KB, 4 packets)
Evidence 2	normal_http.pcapng — baseline capture (10 packets)
Evidence 3	bounded_syn_activity.pcapng — 4-packet simulation (12 packets)
mySYNFloodCapture SHA-256	14765b029a72e9c41dd8b4d32f5b1d2c7d9efee0f084949151f183a39baa55f5
bounded_syn SHA-256	ac88da9466647ae3576017387ef975b05204fab38e0636093cec7b4957166937
Safety Controls	4 packets only, 127.0.0.1 only, stopped immediately after capture

3. Technical Background

3.1 Normal TCP Handshake

SYN (client) → SYN-ACK (server allocates resources) → ACK (connection established) → Data exchange → FIN-ACK (clean closure)

3.2 SYN Flood Mechanism

An attacker sends many SYN packets but never sends the final ACK. Each unanswered SYN-ACK causes the server to maintain a half-open connection entry consuming memory. When the backlog is exhausted, legitimate connections are refused — Denial of Service.

3.3 Key Forensic Indicators

Indicator	Normal Traffic	SYN Flood Pattern
Handshake	SYN → SYN-ACK → ACK	SYN → SYN-ACK → (no ACK)
Source ports	Single consistent port	Many random ports
HTTP data	Present after handshake	Absent
SYN packet size	~74 bytes (no payload)	May contain large payload
RST packets	None	Present if OS rejects SYN-ACK

4. Part A — Normal Handshake Baseline

A normal HTTP session was captured on the loopback interface to establish a comparison baseline.

Frame	Flag	Src Port	Dst Port	Interpretation
1	SYN (0x0002)	60880	80	Client requests TCP connection
2	SYN-ACK (0x0012)	80	60880	Server acknowledges — resources allocated
3	ACK (0x0010)	60880	80	Handshake complete — connection established
4	HTTP GET	60880	80	Normal HTTP request — application data
6	HTTP 200 OK	80	60880	Server response — page served successfully
8	FIN-ACK (0x0011)	60880	80	Client closes connection gracefully
9	FIN-ACK (0x0011)	80	60880	Server closes connection gracefully

Baseline: Complete handshake + HTTP data + clean FIN-ACK closure in under 17ms. This is the expected normal pattern.

5. Part B — Bounded Loopback Simulation

SAFETY: Exactly 4 SYN packets to 127.0.0.1:80 only. No external systems targeted. Stopped immediately after capture.

Scapy script sent 4 SYN packets with random source ports. Result: 12 packets captured — 4 SYN + 4 SYN-ACK + 4 RST.

6. Part C — SYN Indicator Analysis

6.1 Initial SYN Packets (filter: tcp.flags.syn==1 && tcp.flags.ack==0)

Frame	Src Port	Dst Port	Seq	Completed?
1	10980	80	0	No — RST sent
4	12538	80	0	No — RST sent
7	16211	80	0	No — RST sent
10	15770	80	0	No — RST sent

6.2 SYN-ACK Responses — server responded to all 4 SYNs

Frames 2, 5, 8, 11 — server sent SYN-ACK to ports 10980, 12538, 16211, 15770

6.3 RST Packets — OS rejected unsolicited SYN-ACKs

Frames 3, 6, 9, 12 — RST (0x0004) sent by OS. Scapy bypasses the OS TCP stack so the OS had no record of these connections and auto-sent RST. This is a forensic indicator of tool-generated traffic.

7. Part D — Quantitative Analysis

Metric	Value	Forensic Meaning
Total SYN packets	4	Multiple connection attempts without completion
SYN-ACK responses	4	Server consumed resources for each
Completed handshakes	0	Zero legitimate connections established
Unique source ports	4 (10980, 12538, 15770, 16211)	Random ports — tool-generated indicator
RST packets	4	OS-generated — Scapy bypassed kernel stack
Expert WARN events	4 x Connection reset (RST)	Abnormal termination pattern
Expert NOTE events	4 x SYN without SACK PERM	Tool-crafted packets confirmed
Retransmissions	0	Consistent with small-scale simulation
HTTP requests	0	No legitimate application traffic

8. Part E — Normal vs Suspicious Comparison

8.1 Supplied PCAP Analysis

The supplied mySYNFloodCapture.pcap contained 4 SYN packets with 65,000-byte payloads — 877x larger than a normal SYN packet. No SYN-ACK responses were present.

CRITICAL: Normal SYN packets have zero data payload (~74 bytes). These 65,054-byte SYN packets are a strong indicator of tool-generated attack traffic.

8.2 Comparison Table

Indicator	Normal HTTP	Bounded Sim	Supplied PCAP
Initial SYN count	1	4	4
SYN-ACK count	1	4	0
Completed handshakes	1	0	0
Unique source ports	1	4 (random)	4 (random)
HTTP data present	Yes	No	No
RST packets	0	4	0
SYN packet size	~74 bytes	~74 bytes	65,054 bytes
Clean closure	Yes (FIN-ACK)	No	No

9. Forensic Interpretation

9.1 What the Evidence Shows

- Both captures show the SYN flood packet pattern — multiple SYNs with no completed handshakes
- Randomised source ports confirm tool-generated traffic in both captures
- Missing SACK PERM option confirms Scapy-generated packets (not normal OS traffic)
- 65,000-byte SYN payloads in the supplied PCAP indicate a more aggressive tool
- Zero HTTP requests confirm no legitimate web activity occurred

9.2 What the Evidence Does NOT Prove

4 packets alone cannot prove a successful Denial of Service. A defensible DoS conclusion requires: sustained rate/duration data, server log evidence of backlog exhaustion, increased response times or connection refusals, and volume evidence (real floods involve thousands to millions of packets per second).

10. Detection and Mitigation Recommendations

Category	Recommendation
Detection	Alert on elevated SYN-to-completed-handshake ratios
Detection	Monitor SYN backlog exhaustion and retransmission counts
Detection	Threshold alerts for SYN packets from single source per time window
Mitigation	Enable SYN cookies on the server OS
Mitigation	Configure TCP backlog sizes and SYN-ACK retransmission timeouts
Mitigation	Deploy upstream rate limiting and DDoS protection
Logging	Retain full packet capture during incidents
Correlation	Correlate packet evidence with web server and firewall logs

11. Conclusion

This investigation demonstrated forensic analysis of TCP SYN flood patterns. Normal baseline, bounded simulation and supplied PCAP were compared. The SYN flood pattern indicators were clearly identified: incomplete handshakes, random source ports, no HTTP data, and oversized SYN packets. Four training packets reproduce the forensic signature but do not prove service denial without corroborating log evidence.

GitHub: <https://github.com/Adeleye001/SBT-DF203-Lab3-SYNFlood-Investigation>